

Software Quality Assurance

Module 03 — Test Planning & Test Checklist

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Q1. Software Test Plan — Online Shopping Website

This document presents a structured test plan for a hypothetical Online Shopping Website. It defines the scope, approach, resources, and schedule for testing the core e-commerce features of the system. No actual live website has been tested; the platform is treated as a generic, hypothetical system for the purpose of this academic exercise.

1. Test Plan ID / Version

Test Plan ID: TP-OSW-001

Version: 1.0

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2. Introduction / Overview

The Online Shopping Website is a hypothetical e-commerce platform that allows customers to browse products, manage a cart and wishlist, and complete purchases online. This test plan outlines how the quality assurance team will validate that the website's core features function correctly, are user-friendly, and are reasonably secure and reliable before release.

3. Objectives

- Verify that all core e-commerce features work as per the specified requirements.
- Identify and report defects early in the testing cycle to reduce the cost of fixing them.
- Ensure the website provides a consistent and usable experience across common browsers and devices.
- Validate that customer data and payment-related information are handled securely.
- Provide stakeholders with a clear assessment of product quality and release readiness.

4. Scope

4.1 In Scope

- User registration and account creation
- Login and logout functionality
- Product browsing, search, and filtering
- Product detail page
- Shopping cart (add, update, remove items)
- Wishlist management
- Checkout process, including address and payment steps
- Address management (add, edit, delete, set default)
- Payment processing (integration-level, using test/sandbox modes)
- Order placement, order history, and order tracking
- Notifications (email/SMS/in-app, where applicable)

4.2 Out of Scope

- Backend/admin panel functionality (unless separately specified)
- Third-party payment gateway's internal processing logic
- Load testing beyond agreed target user volumes
- Marketing modules such as banner/CMS content management
- Native mobile application testing (web application only, unless stated otherwise)

5. Features to be Tested

- User Registration
- Login / Logout
- Product Search and Filtering
- Product Listing and Product Details
- Shopping Cart
- Wishlist
- Checkout Workflow
- Address Management
- Payment Processing
- Order Placement, History and Tracking
- Notifications

6. Features Not to be Tested

- Internal database schema and stored procedures
- Third-party courier/logistics partner systems
- Payment gateway's internal fraud-detection engine
- Server infrastructure and hosting configuration

7. Test Strategy / Testing Approach

Testing will follow a combination of manual and automated approaches. Manual testing will be used for exploratory, usability, and one-time validation scenarios, while automation will be applied to critical, repeatable flows such as login, add-to-cart, and checkout for regression purposes. Testing will proceed in the order: Smoke Testing → Functional Testing → UI/Usability Testing → Compatibility Testing → Security Testing → Regression Testing → Sanity Testing, aligned with the build and release schedule.

8. Types of Testing

Functional Testing

Validates that each feature (registration, login, cart, checkout, etc.) works according to the specified business requirements.

UI Testing

Verifies layout, alignment, labels, icons, and visual consistency of pages across the website.

Usability Testing

Evaluates ease of navigation, clarity of instructions, and overall user experience for a typical shopper.

Compatibility Testing

Confirms the website behaves consistently across major browsers (Chrome, Firefox, Edge, Safari) and screen sizes (desktop, tablet, mobile).

Performance Testing

Checks page load time, response time under normal load, and behavior of the site during peak browsing/checkout periods.

Security Testing

Validates authentication controls, session handling, input sanitization, and protection of sensitive data such as passwords and payment details.

Regression Testing

Re-executes existing test cases after code changes to confirm that previously working features remain unaffected.

Smoke Testing

Performs a quick check of critical flows (login, add to cart, checkout) after each new build to confirm basic stability.

Sanity Testing

Focuses on verifying that a specific bug fix or minor change works correctly without a full regression cycle.

9. Test Environment

- Staging/QA environment configured to closely mirror production.
- Supported browsers: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari (latest two versions).
- Devices: Desktop, tablet, and mobile (responsive views).
- Payment gateway configured in sandbox/test mode with test card credentials.

10. Test Data

- Test user accounts with varying states (new user, existing user, locked account).
- Sample product catalog covering in-stock, low-stock, and out-of-stock items.
- Valid and invalid coupon/discount codes.
- Test payment credentials provided by the payment gateway's sandbox environment.
- Multiple saved addresses per test account (domestic and edge-case formats).

11. Roles and Responsibilities

Role	Responsibility
QA Lead	Plans the testing effort, reviews test cases, tracks progress, and reports overall quality status.
Test Engineer(s)	Designs and executes test cases, logs defects, and performs regression testing.
Developer(s)	Fixes reported defects and supports root-cause analysis.
Project Manager	Tracks schedule, coordinates between teams, and approves release readiness.
Business Analyst	Clarifies requirements and confirms expected behavior for ambiguous scenarios.

12. Entry Criteria

- Requirement documents and design specifications are available and reviewed
- Test environment is set up and accessible
- Test data is prepared
- Build is deployed to the test environment and smoke testing is passed
- Test cases and checklists are reviewed and approved

13. Exit Criteria

- All planned test cases have been executed
- No open Critical or High severity defects remain unresolved
- At least 95% of test cases have passed
- All Medium/Low severity defects are documented with agreed workarounds or deferrals
- Test summary report is prepared and shared with stakeholders

14. Test Deliverables

- Test Plan document
- Test Cases and Checklists
- Traceability Matrix (requirements to test cases)
- Defect Reports
- Test Execution/Summary Report

15. Defect Management Process

1. Tester identifies an issue and logs it in the defect tracking tool with steps to reproduce, screenshots, severity, and priority.
2. QA Lead reviews and validates the defect, then assigns it to the appropriate developer.
3. Developer analyzes the defect, fixes it, and updates the status to 'Fixed'.
4. Tester re-tests the fix in the test environment and marks it as 'Verified' or reopens it if unresolved.
5. Once verified, the defect is closed. Recurring or related areas are covered through regression testing.

16. Risks and Mitigation

Risk	Mitigation
Unstable or delayed builds	Maintain close coordination with the development team and plan buffer time in the schedule.
Incomplete or changing requirements	Confirm requirements with the Business Analyst before test design and update test cases as needed.
Limited test environment availability	Schedule environment usage in advance and maintain a backup environment if possible.
Third-party payment gateway sandbox downtime	Use mock/stub responses for critical flows when the sandbox is unavailable.
Tight testing timeline before release	Prioritize testing based on risk and business impact (critical flows first).

17. Assumptions and Dependencies

- The application under test is a web-based e-commerce platform accessible via standard browsers.
- A dedicated test environment with test data will be made available before test execution begins.
- Payment processing will be tested using sandbox/test credentials, not real transactions.
- Requirement documents will be finalized before the start of formal test execution.

18. Schedule / Timeline

Activity	Estimated Duration
Test Planning and Test Case Design	3 working days
Test Environment and Data Preparation	1 working day
Smoke Testing	1 working day
Functional and UI Testing	5 working days
Compatibility, Usability and Security Testing	3 working days
Regression Testing	2 working days
Defect Retesting and Sanity Testing	2 working days

Activity	Estimated Duration
Test Closure and Reporting	1 working day

19. Tools & Technologies

Category	Tool(s)
Test Management	Excel / Google Sheets or a test management tool (e.g., TestRail)
Defect Tracking	Jira or an equivalent issue-tracking tool
Automation (optional)	Selenium, Playwright, or Cypress for regression of critical flows
API Testing	Postman
Browser Compatibility	BrowserStack or manual testing on major browsers
Performance Testing (basic)	JMeter

20. Approval / Sign-off

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Reviewed By:

Date: 02/09/2026

Q2. Test Checklists

The following checklists cover Login, Registration, and Checkout for the hypothetical Online Shopping Website. Each checklist contains a minimum of 20 specific, testable validation points, combining positive, negative, boundary, UI, validation, security, and usability scenarios.

A. Login Checklist

SL	Test Area	Validation Point	Expected Result
1	Positive - Valid Login	Enter a valid registered email/username and the correct password, then click the Login button.	User is authenticated successfully and redirected to the homepage/dashboard.
2	Negative - Invalid Email	Enter an email address that is not registered in the system along with any password, then submit.	System displays an error message such as 'Invalid email or password' without confirming which field is wrong.
3	Negative - Invalid Password	Enter a valid registered email with an incorrect password, then submit.	System rejects the login attempt and displays a generic invalid-credentials error message.
4	Validation - Empty Fields	Leave both the email and password fields blank and click the Login button.	System prevents submission and displays 'This field is required' messages for both fields.
5	Validation - Empty Email	Leave the email field blank while entering a valid password, then submit.	System displays a required-field validation message under the email field.
6	Validation - Empty Password	Leave the password field blank while entering a valid email, then submit.	System displays a required-field validation message under the password field.
7	Validation - Email Format	Enter an email address without the '@' symbol (e.g., 'testexample.com') and submit.	System displays an 'Enter a valid email address' validation message.
8	UI - Password Masking	Type a password into the password field and observe how the characters are displayed.	Password characters are masked (shown as dots/asterisks) by default.
9	UI - Show/Hide Password	Click the show/hide password (eye) icon, if available, after typing a password.	Password toggles between masked and plain text view correctly.
10	Functional - Case Sensitivity (Email)	Enter the registered email in a different letter case (e.g., uppercase) with the correct password.	Login succeeds, since email addresses are generally treated as case-insensitive.
11	Security - Case Sensitivity (Password)	Enter the correct email with the correct password typed in a different letter case.	Login fails, since passwords should be treated as case-sensitive.
12	Validation - Leading/Trailing Spaces	Enter a valid email with extra leading or trailing spaces and the correct password, then submit.	System trims the spaces automatically and logs the user in successfully.
13	Functional - Remember Me	Check the 'Remember Me' checkbox, log in successfully, then close and reopen the browser.	User session/credentials are retained and the user remains logged in or the email is pre-filled.

SL	Test Area	Validation Point	Expected Result
14	Functional - Forgot Password Link	Click the 'Forgot Password' link on the login page.	User is redirected to a password-reset page/flow where they can enter their registered email.
15	Functional - Password Reset Flow	Complete the forgot-password flow by entering a registered email and submitting the reset request.	System sends a password reset link/OTP to the registered email and confirms this on screen.
16	Security - Account Lockout	Attempt to log in with an incorrect password multiple times (e.g., 5 consecutive attempts) for the same account.	System temporarily locks the account or displays a warning/CAPTCHA after repeated failed attempts.
17	UI - Error Message Display	Trigger any login validation error and observe where and how the message is displayed.	Error message is clearly visible, appropriately worded, and does not expose sensitive system details.
18	Functional - Login Button State	Observe the Login button while the email/password fields are empty versus filled.	Login button is either disabled until required fields are filled, or clearly clickable with proper validation on click.
19	Functional - Navigation on Success	Log in successfully and observe the page the user is redirected to.	User is redirected to the intended homepage, dashboard, or the page they originally tried to access.
20	Functional - Logout	While logged in, click the Logout button/link from any page.	User session is terminated and the user is redirected to the login or homepage in a logged-out state.
21	Security - Session After Logout	Log out, then click the browser's back button to try to access a previously logged-in page.	System does not display protected content and redirects the user to the login page.
22	Security - Session Timeout	Log in successfully, then remain inactive on the site for the configured session-timeout duration.	System automatically logs the user out and requires re-authentication for further actions.
23	Functional - Multiple Device Login	Log in with the same account from two different browsers/devices simultaneously.	System behaves according to the defined policy (either allows concurrent sessions or invalidates the older session).
24	UI - Field Focus and Tab Order	Use the Tab key to move between the email field, password field, and Login button.	Focus moves in a logical order and the Login button can be triggered using the Enter key.

B. Registration Checklist

SL	Test Area	Validation Point	Expected Result
1	Positive - Valid Registration	Fill in all required fields with valid data (name, email, password, confirm password, etc.) and submit the registration form.	Account is created successfully and a confirmation message or verification prompt is displayed.
2	Validation - Required Fields	Leave all fields empty and click the Register/Sign Up button.	System prevents submission and displays 'This field is required' messages for each mandatory field.
3	Validation - Name Field Empty	Leave the name field blank while filling in all other required fields correctly, then submit.	System displays a required-field error specifically for the name field.
4	Validation - Invalid Name Characters	Enter numbers or special characters (e.g., '12345' or '@#\$%') into the name field.	System rejects the input or displays a validation message indicating only valid name characters are allowed.
5	Validation - Email Format	Enter an improperly formatted email address (e.g., missing '@' or domain) in the email field.	System displays an 'Enter a valid email address' validation message.
6	Functional - Duplicate Email	Attempt to register using an email address that is already associated with an existing account.	System displays a message such as 'This email is already registered' and prevents duplicate account creation.
7	Validation - Password Minimum Length	Enter a password shorter than the minimum required length (e.g., 3 characters) and submit.	System displays a validation message stating the minimum password length requirement.
8	Validation - Password Complexity	Enter a password that does not meet complexity rules (e.g., only lowercase letters, no numbers/symbols) if such rules are enforced.	System displays a message specifying the required password complexity.
9	Validation - Confirm Password Mismatch	Enter a valid password in the Password field and a different value in the Confirm Password field, then submit.	System displays a 'Passwords do not match' error and prevents form submission.
10	Validation - Confirm Password Empty	Enter a valid password but leave the Confirm Password field empty, then submit.	System displays a required-field validation message for the Confirm Password field.
11	UI - Password Masking	Type a password into the Password and Confirm Password fields and observe the display.	Both fields mask the entered characters by default.
12	Validation - Mobile Number Format	Enter an invalid mobile number (e.g., containing letters or an incorrect number of digits) if a mobile field is present.	System displays a validation message indicating the mobile number format is invalid.
13	Functional - Duplicate Mobile Number	Attempt to register using a mobile number that is already linked to an existing account, if uniqueness is enforced.	System displays a message indicating the mobile number is already in use.
14	Validation - Date of Birth (if applicable)	Select or enter a date of birth that indicates the user is below the minimum required age.	System displays a message indicating the user does not meet the minimum age requirement.

SL	Test Area	Validation Point	Expected Result
15	Validation - Terms & Conditions	Leave the 'I agree to the Terms & Conditions' checkbox unchecked and attempt to submit the form.	System prevents registration and prompts the user to accept the terms before proceeding.
16	Functional - CAPTCHA Validation (if applicable)	Attempt to submit the registration form with an incorrect or missing CAPTCHA response.	System rejects the submission and displays a CAPTCHA validation error.
17	Functional - Email/OTP Verification	Complete registration and check whether a verification email or OTP is sent to confirm the account.	System sends a verification email/OTP, and the account is fully activated only after verification is completed.
18	Boundary - Maximum Field Length	Enter an unusually long value (e.g., 300 characters) in the name or email field.	System either restricts input to a defined maximum length or displays an appropriate validation message.
19	Validation - Leading/Trailing Spaces	Enter a valid email or name with extra leading or trailing spaces and submit the form.	System trims unnecessary spaces automatically before validating and saving the data.
20	UI - Error Message Clarity	Trigger any registration validation error and review the message shown to the user.	Error messages are clear, specific to the field, and guide the user on how to correct the input.
21	Functional - Successful Registration Redirect	Complete a valid registration and observe the resulting page or action.	User is redirected to a confirmation page, verification prompt, or automatically logged in, per the defined flow.
22	Security - Password Storage	Register a new account and, from a QA/security perspective, confirm password handling policy with the development team.	Password is not stored or transmitted in plain text; it is handled securely (e.g., hashed) as per design documentation.
23	Functional - Field-Level Real-Time Validation	Enter data into each field one at a time and move focus away (on blur) to check for validation feedback.	System validates each field individually and shows relevant feedback without requiring full form submission.

C. Checkout Checklist

SL	Test Area	Validation Point	Expected Result
1	Positive - Standard Checkout Flow	Add valid items to the cart, proceed to checkout, fill in all required details, and complete the order using a valid payment method.	Order is placed successfully and an order confirmation with a unique Order ID is displayed.
2	Functional - Cart Items Display	Proceed to checkout with multiple items in the cart and verify the listed items on the checkout page.	All items added to the cart are correctly displayed with accurate names, images, and details.
3	Functional - Product Quantity	Increase or decrease the quantity of a product on the checkout/cart page.	Quantity updates correctly and the item subtotal recalculates automatically.
4	Boundary - Minimum Quantity	Attempt to reduce a product's quantity below 1 (e.g., to 0) on the checkout page.	System prevents the quantity from going below 1 or prompts removal of the item from the cart.
5	Functional - Product Price Accuracy	Compare the price shown on the checkout page with the price shown on the product details page.	Price displayed at checkout matches the product's listed price, including any active discount.
6	Functional - Discount/Coupon - Valid Code	Enter a valid, active discount/coupon code and apply it during checkout.	Discount is applied correctly and the total amount is recalculated to reflect the reduced price.
7	Negative - Discount/Coupon - Invalid Code	Enter an invalid, expired, or already-used coupon code and attempt to apply it.	System displays an appropriate error message and does not apply any discount.
8	Functional - Tax Calculation	Review the tax amount calculated on the checkout summary for the items in the cart.	Tax is calculated correctly based on the applicable rate and item/order value.
9	Functional - Shipping Charges	Select different delivery locations or shipping methods and observe the shipping charge.	Shipping charge updates correctly based on the selected location and delivery method.
10	Functional - Total Amount Calculation	Review the final order total, including item subtotal, discount, tax, and shipping charges.	Total amount equals the sum of subtotal minus discount, plus tax and shipping, calculated accurately.
11	Functional - Billing Address Entry	Enter a new billing address with all required fields during checkout.	Billing address is accepted, validated, and saved correctly for the current order.
12	Functional - Shipping Address Entry	Enter a new shipping address, or select a previously saved address, during checkout.	Shipping address is accepted and correctly associated with the order.
13	Validation - Address Field Validation	Leave a mandatory address field (e.g., city or postal code) blank and attempt to proceed.	System displays a required-field validation message and prevents proceeding to the next step.
14	Validation - Invalid Postal Code	Enter an invalid or improperly formatted postal/ZIP code in the address form.	System displays a validation message indicating the postal code format is incorrect.
15	Functional - Multiple Saved Addresses	Select a different address from a list of previously saved addresses during checkout.	Selected address is correctly applied to the current order, replacing any previously chosen address.

SL	Test Area	Validation Point	Expected Result
16	Functional - Delivery Options	Choose between available delivery options (e.g., standard vs. express), if offered.	Estimated delivery date and shipping cost update correctly based on the selected option.
17	Functional - Payment Method Selection	Select each available payment method (e.g., card, mobile banking, cash on delivery) one at a time.	Selected payment method is highlighted/selected and relevant payment fields are displayed accordingly.
18	Negative - Payment Failure Handling	Simulate a failed payment using invalid test card details or a declined sandbox transaction.	System displays a clear payment-failure message and allows the user to retry without losing cart contents.
19	Positive - Payment Success Handling	Complete payment using valid test/sandbox payment details.	Payment is processed successfully and the user is redirected to an order confirmation page.
20	Functional - Order Confirmation Details	Review the order confirmation page/email after successfully placing an order.	Confirmation displays correct order details, including items, total amount, address, and Order ID.
21	Functional - Order ID Generation	Place multiple separate orders and compare the generated Order IDs.	Each order receives a unique, correctly formatted Order ID with no duplicates.
22	Functional - Inventory/Stock Check	Attempt to check out with a product whose available stock is less than the quantity requested.	System displays a message indicating limited stock and adjusts or restricts the quantity accordingly.
23	Negative - Out-of-Stock Product	Attempt to proceed to checkout with a product that has become out-of-stock after being added to the cart.	System notifies the user that the item is out of stock and prevents checkout until it is removed.
24	Security - Session Timeout During Checkout	Remain idle on the checkout page beyond the configured session-timeout duration, then attempt to complete the order.	System prompts re-authentication or displays a session-expired message, without processing an invalid order.
25	Functional - Duplicate Order Prevention	Click the 'Place Order' or 'Pay Now' button multiple times in quick succession.	System processes only a single order and prevents duplicate order/payment submission.